

SENA YALÇIN

Junior Computer Engineer

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Education

Hacettepe University

September 2020 - June 2025

Bachelor of Science in Computer Science (GPA: 3.29 / 4.00)

Ankara, Turkey

- **Relevant Coursework:** Machine Learning (PyTorch, TensorFlow), Computer Vision (OpenCV), AI Algorithms, Computer Networks (Wireshark, Cisco CLI), Prob & Stat in CS (Python), Algorithms (Java), Software Documentation (Signavio), Software Engineering (Spring Boot, React.js), Data Structures (C++), Data Management (MySQL, PostgreSQL), Secure Programming (Linux), Introduction to Blockchain, Introduction to Logic (Verilog)

Experience

EHSİM A.Ş.

June 2024 – July 2024

Software Engineering Intern

Ankara, Turkey

- Designed and implemented algorithms for missile threat analysis, processing trajectory, altitude, and velocity data to automate countermeasure selection.
- Developed a C# based desktop application simulating F-16 defense strategies, integrating flight dynamics and real-time monitoring of RF and IR-guided threats.
- Engineered a graphical user interface (GUI) for real-time threat visualization, incorporating manual override controls for countermeasure deployment.
- Conducted comprehensive testing using randomized datasets and edge-case scenarios, ensuring robust performance and accurate simulation under varied conditions.

Garanti BBVA Teknoloji

July 2024 – October 2024

Software Engineering Intern

Istanbul, Turkey

- Automated testing for over 30 critical RESTful APIs using the Karate Framework, ensuring system functionality for banking operations such as user authentication and transaction processing.
- Optimized legacy COBOL code by refactoring loops, improving SQL query performance, and restructuring database access patterns, resulting in a 15% reduction in CPU usage and a 20% improvement in transaction processing speeds.
- Integrated automated test suites into Jenkins pipelines, streamlining continuous integration workflows and reducing regression testing cycles by 40%.

EHSİM A.Ş.

November 2024 – Present

Software Engineer

Ankara, Turkey

- Contributing to the development of software components for radar simulation systems used in electronic warfare and defense applications.
- Developing simulation and analysis tools using Java, C#, and MATLAB for modeling radar behavior and system interactions.
- Conducting exploratory studies on the application of artificial intelligence (AI) techniques within radar related software systems.
- Completed a company provided professional C++ training program, gaining experience with modern C++ standards and best practices.
- Supporting version control and branching strategies (GitFlow) within CI/CD processes.

Technical Skills

Programming Languages: Python, Java, C#, C++

DBMS: MySQL, PostgreSQL

Technologies and Frameworks: PyTorch, TensorFlow, Spring Boot, React.js, JavaFX

Operating Systems: Linux, Windows

DevOps & CI/CD: Git, GitLab CI/CD, Bitbucket Pipelines

Projects

DeepFake Detection Model Development

Technologies: Python, PyTorch, TensorFlow

Guided by Prof. Nazlı İKİZLER CİNBIŞ

- Leading research on deepfake detection, with an emphasis on real-world applications, particularly in political contexts, to combat misinformation and ensure content integrity.
- Developing and training advanced models, with the goal of creating a custom deepfake detection solution and a deepfake detection tool aimed at preventing the societal impacts of manipulated media.
- Currently writing a research paper that will be published in academic journals, detailing our methodologies, findings, and contributions.

Surveillance Image Recognition Research Project

Technologies: Python, PyTorch, OpenCV

- Researched methods to enhance face recognition accuracy on low-resolution surveillance images using state-of-the-art super-resolution (SR) techniques.
- Implemented and evaluated multiple SR models, including Bicubic, ESR-CNN, LapSRN, ESPCN, Real-ESRGAN, GPEN, and SwinIR, to improve image quality prior to recognition.
- Integrated MTCNN and InceptionResNetV1 for face recognition and achieved a 32% improvement in accuracy through SR-based preprocessing.
- Currently writing a research paper that will be published in academic journals, detailing our methodologies, findings, and contributions.

IMECE Satellite Path Optimization Algorithm Development

Technologies: Java, RegEx

Guided by Asst. Prof. Selma DİLEK

- Designed and implemented algorithms to optimize pathfinding for a surveillance drone on TÜBİTAK's IMECE satellite project, considering constraints like fuel cost, elevation, and maximum height.
- Developed efficient methods for parsing input data using regular expressions and applied shortest-path and greedy programming strategies.
- Generated grayscale topographic maps for visualization and provided cost-efficient solutions for mission-critical drone operations.

Review and Recommendation Website

Technologies: Java, React.js

- Developed a review and recommendation platform with a backend built in Java. Integrated user authentication, ratings, and recommendations.
- Implemented machine learning algorithms to generate personalized recommendations based on user behavior and ratings. Created RESTful APIs to handle user data and review functionalities.

Professional Affiliations and Activities

- Member, Association for Computing Machinery (ACM)
- Member, Google Developer Students Club (GDSC Hacettepe)
- Member, Institute of Electrical and Electronics Engineers (IEEE)

Additional Skills

Languages: English (C1), Turkish (C2), German (B1)

Certificates: Google Cloud Machine Learning Bootcamp, Global AI Hub - Akbank Python Bootcamp, Global AI Hub - Aygaz AI Literacy Bootcamp, Patika.dev Data Science Certificate

Soft Skills: Problem Solving, Self-Motivated, Training and Development, Leadership